

Worked Solutions

Data Analytics

Set B: Original | CSE Semester 3

Companion to da-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Arithmetic mean

Correct option: (C). The sum is 48 and $48/6 = 8$.

2. Median

Correct option: (B). The middle values are 6 and 8, so the median is $(6 + 8)/2 = 7$.

3. Population variance

Correct option: (A). Population variance divides the squared-deviation sum by N : $64/6$.

4. Pearson skewness sign

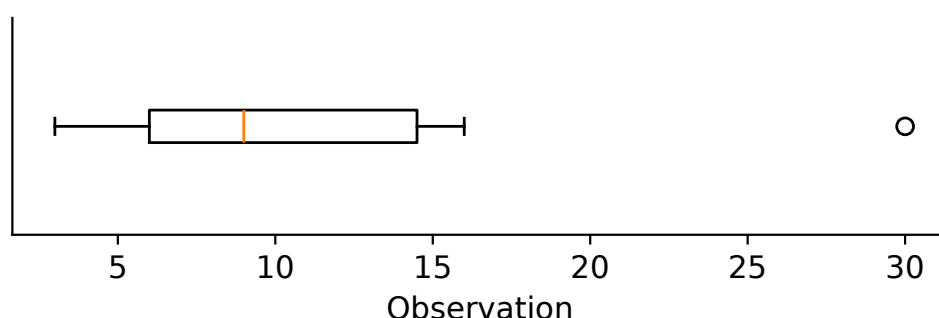
Correct option: (D). $3(\bar{x} - \text{median})/\sigma = 3(8 - 7)/\sigma > 0$.

5. Resistance to extremes

Correct option: (B). The median depends on the middle ranks and is generally more resistant to extremes.

6. Five-number summary and outlier

The median is 9. Excluding it gives lower half (3, 5, 7, 8) and upper half (11, 13, 16, 30). Thus $Q_1 = 6$, $Q_3 = 14.5$, and the five-number summary is (3, 6, 9, 14.5, 30). $IQR = 8.5$; the fences are -6.75 and 27.25 . Therefore 30 is an outlier. The modified box plot has whiskers at 3 and 16, with 30 plotted separately.



7. Missing and inconsistent data

Impute missing ages with the median of a relevant, sufficiently large customer group and retain a missingness flag. Estimate imputation values from training data only if building a predictive model. Flag rows where annual income differs materially from $12 \times$ monthly income after aligning currency, units, gross/net definitions and time period. Investigate such rows rather than automatically overwriting them.

8. Filling-machine mean test

$H_0 : \mu = 500$ ml, $H_1 : \mu \neq 500$ ml. The standard error is $12/\sqrt{36} = 2$ ml.

$$z = \frac{496 - 500}{2} = \boxed{-2.00}.$$

Since $|-2| > 1.96$, reject H_0 at 5%. There is evidence that the machine's mean fill differs from 500 ml; the sample suggests underfilling. The two-sided p -value is about 0.0455.

9. Rank correlation

The six rank differences are $(-1, 1, -1, 1, -1, 1)$, giving $\sum d_i^2 = 6$.

$$r_s = 1 - \frac{6(6)}{6(6^2 - 1)} = \boxed{\frac{29}{35} = 0.8286}.$$

There is strong positive rank agreement, although each adjacent pair is reversed.

10. Equal-frequency smoothing

Bin	Observations	Mean
1	10,12,15,18	$55/4 = 13.75$
2	20,22,25,28	$95/4 = 23.75$
3	30,32,35,40	$137/4 = 34.25$

The smoothed sequence is four copies of 13.75, followed by four copies of 23.75 and four copies of 34.25.

11. Training and passing

H_0 : training and passing are independent. Row totals are 60 and 40; column totals are 70 and 30. Expected counts are

$$E = \begin{pmatrix} 42 & 18 \\ 28 & 12 \end{pmatrix}.$$

$$\chi^2 = \frac{(48 - 42)^2}{42} + \frac{(12 - 18)^2}{18} + \frac{(22 - 28)^2}{28} + \frac{(18 - 12)^2}{12} = \boxed{7.1429}.$$

With one degree of freedom, $7.1429 > 3.841$, so reject independence. Training and passing are associated. The pass rates are 80% and 55%; causation requires an appropriate study design.

12. One-way ANOVA

Source	SS	df	MS	F
Between	54	2	27	27
Within	6	6	1	
Total	60	8		

Test $H_0 : \mu_A = \mu_B = \mu_C$. Since $F = 27 > 5.143$, reject H_0 at 5%. At least one population mean differs. Pairwise differences require further testing.

13. Type I and Type II errors

A Type I error is rejecting $\mu = 500$ when the machine's true mean is 500 ml. A Type II error is failing to reject $\mu = 500$ when the true mean differs from 500 ml. The 5% significance level controls the Type I error probability under the test assumptions; it does not specify the Type II probability.

Worked Solutions

Data Structures

Set B: Original | CSE Semester 3

Companion to ds-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Structure definition

Correct option: (C). `struct` defines a type whose members can hold different kinds of data.

2. Nested member access

Correct option: (A). For structure objects, each member is accessed with a dot: `s.admitted.year`.

3. String capacity

Correct option: (D). One of the 40 positions must hold the terminating null character.

4. Safe input width

Correct option: (B). `%39s` reads at most 39 characters and appends the null terminator.

5. Pointer member access

Correct option: (C). `p->roll` is equivalent to `(*p).roll`.

6. Contiguous dynamic matrix

```
#include <stdlib.h>
int *makeMatrix(int r, int c) {
    if (r <= 0 || c <= 0) return NULL;
    /* The question guarantees that this size fits size_t. */
    return malloc((size_t)r * (size_t)c * sizeof(int));
}
/* Access (i,j) as *(base + (size_t)i * c + j).
Valid indices satisfy 0 <= i < r and 0 <= j < c.
Caller checks the returned pointer and eventually frees it. */
```

Rows are adjacent in memory, so row i starts at offset ic and column j adds j . Allocation requests $rc \text{ sizeof(int)}$ bytes. A non-null result must be initialized before its elements are read and released with `free` when no longer needed.

7. Dangling pointer and memory leak

A dangling pointer refers to storage whose lifetime has ended:

```
int *p = malloc(sizeof *p);
free(p);      /* p now dangles; do not dereference it. */
p = NULL;     /* Clear this pointer after releasing storage. */
```

Any aliases must also stop using that storage. A memory leak loses the final reference to allocated storage:

```
int *p = malloc(sizeof *p);
p = NULL;     /* Wrong: allocated storage can no longer be freed. */
```

Keep an owning pointer and free the allocation before overwriting that final reference. These examples illustrate errors; they are not code to copy into a working implementation.

8. Delete all matching nodes

The pointer link always addresses the pointer that leads to the current node. On deletion, update that pointer and keep link in place so the next node is checked. On retention, advance to the next link. This covers head, consecutive and all-node matches, and releases each removed allocation. Time is $O(n)$; auxiliary space is $O(1)$. The following shared listing also contains the reversal for 9.

```
#include <stdlib.h>
typedef struct Node {
    int value;
    struct Node *next;
} Node;
void deleteAll(Node **head, int key) {
    if (!head) return;
    Node **link = head;
    while (*link) {
        if ((*link)->value == key) {
            Node *old = *link;
            *link = old->next;
            free(old);
        } else link = &(*link)->next;
    }
}
Node *reverse(Node *head) {
    Node *prev = NULL, *cur = head;
    while (cur) {
        Node *next = cur->next;
        cur->next = prev;
        prev = cur;
        cur = next;
    }
    return prev;
}
```

9. Reverse a singly linked list

Use the reverse function in the listing for 8. Save the original successor before changing cur's next pointer; move prev and cur forward. The loop invariant is that prev heads the reversed processed prefix and cur heads the untouched suffix. At termination, prev is the new head. Assign `head = reverse(head)`. Empty and one-node lists work unchanged. Time is $O(n)$ and auxiliary space is $O(1)$.

10. List trace

$$\begin{array}{ccccccc}
 10 \rightarrow 20 \rightarrow 30 & \longrightarrow & 5 \rightarrow 10 \rightarrow 20 \rightarrow 30 \\
 \longrightarrow & 5 \rightarrow 10 \rightarrow 20 \rightarrow 25 \rightarrow 30 & \longrightarrow & \boxed{5 \rightarrow 10 \rightarrow 20 \rightarrow 25}.
 \end{array}$$

Starting at the final head, follow three next links to reach tail 25. Checking that the tail's next pointer is NULL does not traverse another node-to-node link.

11. Stack trace for infix conversion

The stack is listed bottom to top. Exponentiation is right-associative, so the second exponentiation does not pop the first.

Token	Operator stack	Output
A	empty	A
+	+	A
B	+	AB
*	+ *	AB
(	+ * (	AB
C	+ * (	ABC
-	+ * (-	ABC
D	+ * (-	ABCD
)	+ *	ABCD-
^	+ * ^	ABCD-
E	+ * ^	ABCD-E
^	+ * ^ ^	ABCD-E
F	+ * ^ ^	ABCD-EF
End	empty	ABCD-EF^^*+

Thus the final postfix form is `ABCD-EF^^*+`. The two exponentiations encode $(C - D)^{(E^F)}$.

12. Balanced brackets

Push each opening bracket. For a closing bracket, require a nonempty stack and matching opening type. Accept only if the stack is empty at the end. Time is $O(n)$; the fixed array uses capacity 100, as permitted by the question.

```
int balanced(const char *s) {
    char stack[100];
    int top = 0;
    for (; *s; ++s) {
        if (*s == '(' || *s == '[') {
            if (top == 100) return 0;
            stack[top++] = *s;
        } else if (*s == ')' || *s == ']') {
            if (!top) return 0;
            char open = stack[--top];
            if ((*s == ')' && open != '(') ||
                (*s == ']' && open != '[')) return 0;
        } else return 0;
    }
    return top == 0;
}
```

13. Undo/redo trace

Operation	Undo: bottom to top	Redo: bottom to top
Edits A, B, C	A, B, C	empty
Undo	A, B	C
Undo	A	C, B
Redo	A, B	C
New edit D	A, B, D	empty

Undo transfers the newest applied action to redo; redo reapplies its newest undone action. New edit D clears redo, since C belongs to the abandoned editing branch.

Worked Solutions

Data Communication and Computer Networks

Set B: Original | CSE Semester 3

Companion to dccn-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Nyquist rate

Correct option: (C). $R = 2B \log_2 L = 2(3000) \log_2 4 = 12000$ bps.

2. Bits per symbol

Correct option: (B). Four levels represent $\log_2 4 = 2$ bits per symbol.

3. Shannon capacity

Correct option: (A). $C = B \log_2(1 + S/N) = 3000 \log_2 32 = 15000$ bps.

4. Tighter rate bound

Correct option: (D). Both limits must hold, so the smaller upper bound, 12 kbps, is tighter.

5. Nyquist bandwidth change

Correct option: (C). $R = 2B \log_2 L$ is directly proportional to B when L is fixed.

6. Modulation comparison

ASK changes carrier amplitude, FSK changes carrier frequency, and PSK changes carrier phase. Binary FSK commonly uses two separated frequencies and needs more bandwidth than binary PSK under comparable simple pulse shaping. ASK and BPSK can have comparable nominal bandwidth for the same bit rate and shaping; exact occupied bandwidth depends on the implementation. ASK is directly vulnerable to amplitude disturbances. Constant-envelope FSK and PSK are less affected by such disturbances, although phase/frequency noise and receiver synchronization still matter.

For a coherent receiver in an approximately additive-noise, bandwidth-limited telephone channel, BPSK is a defensible choice: it uses antipodal signals, offers good bit-error performance at a given energy per bit, and avoids the tone separation of binary FSK. A noncoherent, very simple receiver could instead favor FSK. The noise model and implementation must be stated rather than claiming one scheme is always best.

7. Header overhead

Header bytes = $5(20) = 100$. Total bytes = $900 + 100 = 1000$. Header percentage = $100/1000 \times 100 = 10\%$. Payload efficiency is 90%.

8. CRC division and receiver check

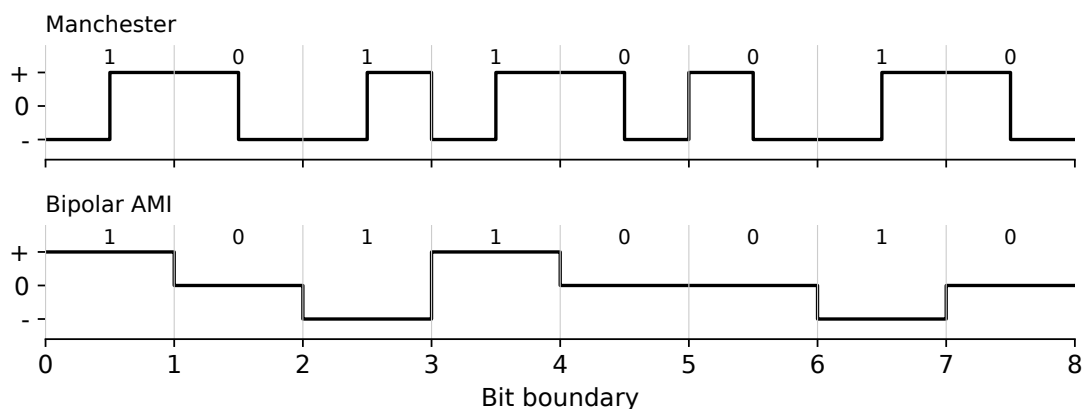
Append four zeros because the generator 10011 has degree four. The aligned XOR working dividends are:

```
11010110110000  initial dividend
01001110110000
00000010110000
00000000101000
00000000001110
```

The remainder is **1110** and the transmitted codeword is **11010110111110**. The receiver divides the whole received codeword by 10011. A zero remainder means no error was detected; it does not guarantee that no undetectable error occurred.

9. Manchester and AMI waveforms

Manchester half-bit pairs are $(-, +), (+, -), (-, +), (-, +), (+, -), (+, -), (-, +), (+, -)$. AMI levels are $+, 0, -, +, 0, 0, -, 0$. These use the conventions printed in the question.



10. Hub and learning switch

A hub repeats the incoming physical signal on every other port. A learning switch records source MAC to ingress-port mappings. For a known destination on a different port, it forwards only there; for a known destination on the incoming port, it filters the frame. For an unknown destination it floods on other forwarding ports in the same VLAN. Broadcasts are also flooded within that VLAN.

11. Stop-and-wait throughput

Frame length $L = 1000(8) = 8000$ bits. Serialization time:

$$t_t = 8000/10^6 = \boxed{8 \text{ ms}}.$$

Cycle time from starting a frame until its ACK returns:

$$t_{\text{cycle}} = t_t + 2t_p = 8 + 20 = \boxed{28 \text{ ms}}.$$

$$U = \frac{8}{28} = \boxed{0.285714}, \quad \text{throughput} = \frac{8000}{0.028} = \boxed{285714.3 \text{ bps}}.$$

Thus utilization is 28.57% and useful throughput is approximately 285.7 kbps. There are no headers or losses in this calculation.

12. Selective repeat versus go-back-N

Selective repeat accepts frame 0 and buffers frames 2 and 3 while waiting for 1. Only frame 1 remains unacknowledged and is retransmitted. After it arrives, the receiver delivers 1, 2 and 3, giving overall delivery order 0, 1, 2, 3. There are five data transmissions in total.

Go-back-N accepts 0, discards 2 and 3, and uses cumulative acknowledgements. After frame 1 times out, it retransmits the outstanding suffix 1, 2 and 3. There are seven data transmissions in total and the same final delivery order.

13. Minimum collision-detection frame

$$L_{\min} = R(2t_p) = 100 \times 10^6 \times 10 \times 10^{-6} = \boxed{1000 \text{ bits} = 125 \text{ bytes}}.$$

This is the propagation-only lower bound for the hypothetical link in the question.

Worked Solutions

Digital Systems and Computer Organization

Set B: Original | CSE Semester 3

Companion to dsco-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. K-map cell count

Correct option: (C). There is one cell per input combination, so $2^4 = 16$ cells.

2. K-map adjacency

Correct option: (A). Gray-code ordering makes adjacent cells differ in exactly one variable.

3. SOP grouping

Correct option: (D). Groups contain a power of two cells; 8 is the only power of two listed.

4. Function reduction

Correct option: (B). Every listed minterm has $B = D$. Thus $F = \neg B \neg D + BD$, independent of A and C.

5. NAND inverter

Correct option: (C). Connecting both inputs to X gives $\neg(X \cdot X) = \neg X$.

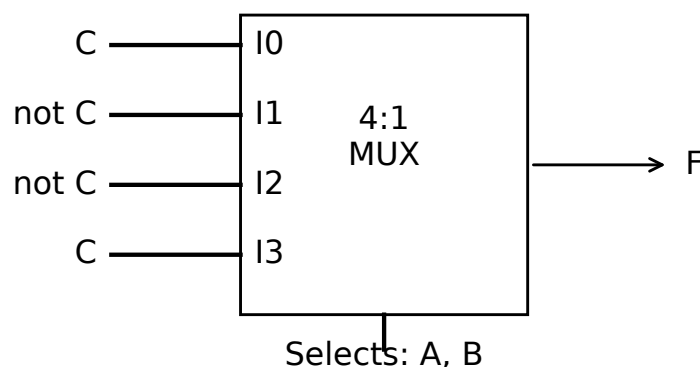
6. Signed results and overflow

Operation	Binary addition	Result	Interpretation	Overflow
11+7	01011 + 00111	10010	-14	Yes
-9+5	10111 + 00101	11100	-4	No
6-13	00110 + 10011	11001	-7	No

Only the true result 18 is outside $[-16, 15]$. For subtraction, 13 is 01101 and its two's-complement negative is 10011. A carry out by itself is not the signed-overflow flag.

7. MUX realization

With A and B as select bits (A most significant), $AB = 00, 01, 10, 11$ produces $C, \bar{C}, \bar{C}, C$, respectively. Thus $I_0 = C, I_1 = \bar{C}, I_2 = \bar{C}, I_3 = C$. One inverter supplies both complemented-C inputs.



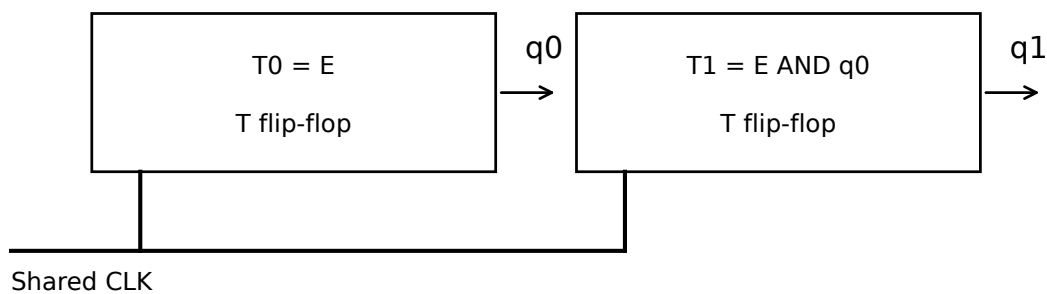
8. Enabled two-bit synchronous counter

Present q_1q_0	Next when E=0	Next when E=1
00	00	01
01	01	10
10	10	11
11	11	00

Use the T excitation equation $T_i = q_i \oplus q_i^+$. The low bit toggles exactly when E=1; the high bit toggles exactly when E=1 and the current low bit is 1.

$$T_0 = E, \quad T_1 = E q_0.$$

Feed E directly to T_0 and an AND of E and q_0 to T_1 . The clock is common.



9. Four-bit ring counter

```

module ring4(input wire clk, input wire reset,
             output reg [3:0] q);
    always @(posedge clk) begin
        if (reset) q <= 4'b0001;
        else q <= {q[2:0], q[3]};
    end
endmodule

```

On a positive clock edge with reset high, q becomes 0001. With reset low, successive states are 0010, 0100, 1000 and 0001. Concatenation moves bits 2:0 upward and puts old bit 3 into bit 0. Nonblocking assignments model sequential storage.

10. Ripple delay and decoded glitches

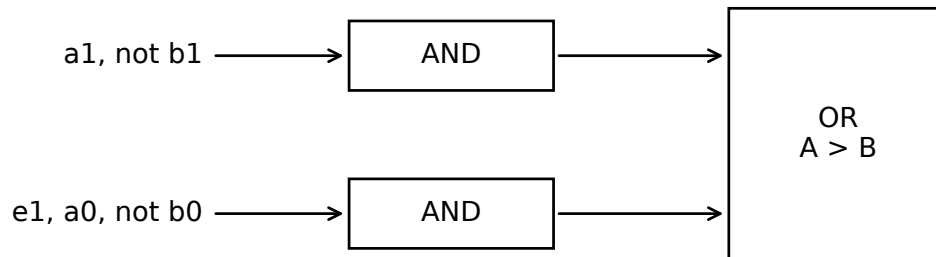
A ripple counter clocks stage 0 from the external clock and later stages from preceding flip-flop outputs. Propagation delays accumulate, so several output bits do not change simultaneously. During a transition such as 0111 to 1000, a decoder may briefly recognize an intermediate pattern. A synchronous counter drives all flip-flops from one shared clock and computes excitation logic from the old state. It avoids stage-by-stage clock ripple, although real output skew and combinational delays still require proper timing design.

11. Two-bit magnitude comparator

Let $e_i = a_i \text{ XNOR } b_i$. Then

$$G = a_1 \bar{b}_1 + e_1 a_0 \bar{b}_0, \quad E_q = e_1 e_0, \quad L = \bar{a}_1 b_1 + e_1 \bar{a}_0 b_0.$$

The upper bit decides unless equal; only then does the lower bit decide. Use XNOR gates for e_1, e_0 , inverters for complements, AND gates for product terms and OR gates to combine them. The figure gives the G network. Swap a and b for L, and AND the two XNOR outputs for E_q .



For A=10 and B=01, $e_1 = 0$, $a_1 \bar{b}_1 = 1$, so $G = 1, E_q = 0, L = 0$.

12. Memory capacity and addresses

4 KiB = 4096 bytes. Each 32-bit word occupies 4 bytes:

$$\text{words} = 4096/4 = 1024, \quad \text{byte-select bits} = \log_2(4096) = 12.$$

The word at 0x1000 occupies 0x1000, 0x1001, 0x1002 and 0x1003. The next word starts at 0x1004. The 12 bits select an offset within the 4 KiB block; the full system may use more address bits to place that block at base 0x1000. Endianness changes byte significance, not these occupied addresses.

13. Memory read and write

For a read: place the target address in MAR, assert the read control signal, wait until the memory interface indicates valid data, and latch the returned value into MDR. Transfer MDR to the destination register as required.

For a write: place the target address in MAR and outgoing data in MDR, assert write with the required timing, and wait for completion. Read moves data from memory to MDR; write moves data from MDR to memory. MAR selects the location in both cases.

Worked Solutions

Discrete Mathematical Structures

Set B: Original | CSE Semester 3

Companion to dms-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Bottom of divisor lattice

Correct option: (C). 1 divides every element of this set.

2. Top of divisor lattice

Correct option: (B). Every element of the set divides 12.

3. Meet of divisors

Correct option: (A). The meet is the greatest common divisor, $\gcd(4, 6) = 2$.

4. Join of divisors

Correct option: (D). The join is the least common multiple, $\text{lcm}(4, 6) = 12$.

5. Complement existence

Correct option: (C). A complement c needs $\gcd(2, c) = 1$ and $\text{lcm}(2, c) = 12$. The only odd divisors, 1 and 3, give lcms 2 and 6; neither works.

6. Truth table and canonical forms

x	y	z	$\neg y$	$x \wedge \neg y$	F
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

Thus $F = \Sigma m(1, 3, 4, 5, 7) = \Pi M(0, 2, 6)$.

Canonical DNF: $F = (\neg x \wedge \neg y \wedge z) \vee (\neg x \wedge y \wedge z) \vee (x \wedge \neg y \wedge \neg z) \vee (x \wedge \neg y \wedge z) \vee (x \wedge y \wedge z)$.

Canonical CNF: $F = (x \vee y \vee z) \wedge (x \vee \neg y \vee z) \wedge (\neg x \vee \neg y \vee z)$.

7. Boolean identity proof

$$\begin{aligned}
 a \vee (\neg a \wedge b) &= (a \vee \neg a) \wedge (a \vee b) && \text{distributive law} \\
 &= 1 \wedge (a \vee b) && \text{complement law} \\
 &= a \vee b && \text{identity law.}
 \end{aligned}$$

8. Integer solutions by inclusion-exclusion

Set $u = x - 1$ and $v = y - 2$. Then $u + v + z = 9$, with $0 \leq u, v \leq 4$ and $z \geq 0$. The unrestricted count is $\binom{11}{2} = 55$. Violations $u \geq 5$ and $v \geq 5$ each leave a nonnegative sum of 4 and each contribute $\binom{6}{2} = 15$. Both cannot occur simultaneously because their sum would be at least 10. Therefore

$$55 - 15 - 15 = 25.$$

As a check, each of the five possible x values can pair with each of the five possible y values; even the maximum sum is $5 + 6 = 11 \leq 12$, and each pair fixes a nonnegative z .

9. 42nd lexicographic permutation

Use zero-based rank $42 - 1 = 41$. Factorial block choices are:

Available symbols	Block size	Quotient / remainder	Chosen
0,1,2,3,4	24	$41 = 1(24) + 17$	1
0,2,3,4	6	$17 = 2(6) + 5$	3
0,2,4	2	$5 = 2(2) + 1$	4
0,2	1	$1 = 1(1) + 0$	2
0	1	$0 = 0(1) + 0$	0

Choose the quotient-indexed symbol from the sorted remaining list at each step, with zero-based indices. The result is $\boxed{13420}$.

10. Arrangements of BALLOON

There are 7 letters, with L repeated twice and O repeated twice. The number of distinct arrangements is

$$\frac{7!}{2!2!} = 1260.$$

If the two Ls must be adjacent, treat LL as one object. The six objects are LL, B, A, O, O, N, so the count is

$$\frac{6!}{2!} = 360.$$

There is no additional factor for exchanging the identical Ls within their block.

11. Dijkstra on the undirected graph

Initialize A to 0 and every other tentative distance to infinity. All weights are nonnegative. After each settlement, relax each incident edge.

Settled	A	B	C	D	E
Initial	0	∞	∞	∞	∞
A	0	4	1	∞	∞
C	0	3	1	6	9
B	0	3	1	4	9
D	0	3	1	4	7
E	0	3	1	4	7

Destination	Shortest path	Distance
A	A	0
B	$A \rightarrow C \rightarrow B$	3
C	$A \rightarrow C$	1
D	$A \rightarrow C \rightarrow B \rightarrow D$	4
E	$A \rightarrow C \rightarrow B \rightarrow D \rightarrow E$	7

12. Connectivity proof

Suppose the graph is disconnected. If a component has s vertices, every vertex in it has degree at most $s - 1$. The minimum-degree assumption forces

$$s \geq \delta(G) + 1 \geq (n + 1)/2.$$

There are at least two components, so their combined size would be at least $n + 1$, contradicting the total of n vertices. Hence the graph is connected.

13. Degree and congruence

For a nonempty simple r -regular graph, its complement is $(n - 1 - r)$ -regular. Self-complementarity gives $r = n - 1 - r$, hence $r = (n - 1)/2$. Thus n is odd. By the handshake lemma, nr is even, so r must be even because n is odd. Writing $r = 2t$ gives $n = 4t + 1$, or $n \equiv 1 \pmod{4}$.